



Hearing Protector Use & Care

Help your employees make the most of the hearing protection devices (HPDs) they wear by verifying that the noise reduction provided is adequate and that the HPDs are properly used, maintained and replaced.

Determining the Adequacy of Hearing Protectors

The Noise Reduction Rating (NRR) is a description of how much noise reduction was measured on a group of well-trained hearing protection users who wore the HPD correctly in a laboratory test. As the science has evolved in hearing protection, it is becoming generally recognized that the best way to ensure the adequacy of a hearing protector for any specific worker is to the conduct fit testing to confirm that the expected level of attenuation is achieved (i.e., the proper type of protector was selected and is being used correctly). 3M strongly recommends the fit testing of hearing protectors. Although U.S. regulations do not yet address fit testing, OSHA regulation 29 CFR 1910.95 specifies that the NRR may be used to determine whether a particular hearing protector provides adequate protection within a given noise exposure in one of the following ways:



- Subtract 7 dB from the NRR, then subtract that value from the A-weighted TWA of the employee, or
- Subtract the Noise Reduction Rating (NRR) from the C-weighted TWA of the employee, or
- Use one of several more in-depth methods developed by NIOSH. These are not widely used.

If your company uses A-weighting to measure the TWA noise exposure of employees, the OSHA method for determining adequacy of hearing protection using the NRR is shown below.

Examples of How NRR is Used

In the U.S., OSHA has provided guidance on how employers are to use the NRR to estimate the noise reduction obtained by employees. These examples are based on information in the [OSHA Technical Manual section III, chapter 5, appendix E](#).

Example 1:
 OSHA 7 dB correction
 TWA is 95 dBA
 NRR is 31 dB

Protected exposure =
 $TWA - (NRR - 7)$
 $95 \text{ dBA} - (31 - 7)$
 $95 \text{ dBA} - 24 \text{ dB}$
 71 dBA

Example 2:
 OSHA 50% derating
 TWA is 95 dBA
 NRR is 31 dB

Protected exposure =
 $TWA - [0.5 \times (NRR - 7)]$
 $95 \text{ dBA} - [0.5 \times (31 - 7)]$
 $95 \text{ dBA} - [0.5 \times 24]$
 $95 \text{ dBA} - 12$
 83 dBA

Example 3:
 OSHA 5 dB adjustment
 for dual protection
 TWA is 95 dBA
 NRR is 31 dB

Protected exposure =
 $TWA - [0.5 \times (NRR - 7) + 5]$
 $95 \text{ dBA} - [0.5 \times (31 - 7) + 5 \text{ dB}]$
 $95 \text{ dBA} - [0.5 \times 24 + 5 \text{ dB}]$
 $95 \text{ dBA} - [12 + 5 \text{ dB}]$
 78 dBA

7 dB Correction

If your company uses A-weighting to measure the TWA noise exposure of employees, the OSHA method for determining adequacy of hearing protection using the NRR is shown in Example 1 above.

In Example 1, the protected exposure for a group of employees who wear this hearing protector is estimated to be 71 dBA. This protected exposure level is well below the OSHA PEL of 90 dB TWA so the hearing protector attenuation would be considered adequate for this exposure.

NOTE: When C-weighting is used to measure the TWA of employees, OSHA does not require use of the 7 dB correction to the NRR.

Derating

The practice of reducing the NRR value, or derating, is done to account for the difference between the NRR and the attenuation workers get when on the job. Since the 1980s, OSHA enforcement policy in the U.S. has been to apply a 50% derating to the NRR as a safety factor when employers use hearing protectors as a substitute for noise controls. See Example 2 to learn how to apply the OSHA 50% derating.

In Example 2 above, the protected exposure for a group of employees who wear this hearing protector is estimated to be 83 dBA. This protected exposure level is below the OSHA PEL of 90 dB TWA so the employer who uses this hearing protector as a substitute for noise controls would likely be considered to be in compliance with OSHA noise control requirements.

Dual Protection

Wearing earmuffs together with earplugs provides between 5-10 dB more noise reduction than either device worn by itself. The extra protection provided varies depending on the frequency of the sound.

In the U.S., OSHA allows employers to add 5 dB to whichever device has the highest NRR to estimate the combined protection when dual protection is worn. In most cases, this 5 dB adjustment is factored in after the NRR has been corrected for A-weighting and derating has been applied. See Example 3 above.

Use and Care of Hearing Protectors

Always follow the User Instructions provided with the hearing protectors for fitting, cleaning & storage, and replacement of HPDs. In general, replace hearing protectors when they are damaged or no longer provide an effective noise-blocking seal.



Disposable foam earplugs

Uses: Several

Care: Not washable. Replace when dirty, damaged or no longer regain original shape after rolling

[See our Disposable Foam Earplugs](#)



Push-to-fit foam earplugs

Uses: Several

Care: Not washable. Replace when dirty, tips detach from stem or tips are no longer soft and pliable

[See our Push-to-fit Foam Earplugs](#)



Reusable earplugs

Uses: Many

Care: Washable. Replace when flanges are damaged, torn or are no longer soft and pliable

[See our Reusable Earplugs](#)



Earmuffs

Uses: Many

Care: Headband and outside of cups are washable. Foam inserts inside cups are not. Replace bands when: damaged or they no longer provide enough tension to hold cups tightly over ears. Replace cushions and foam inserts: every six months or sooner if they are damaged or are no longer soft and pliable.

[See our 3M™ PELTOR™ Earmuffs](#)



Banded hearing protectors

Uses: Many

Care: Bands and reusable tips are washable. Foam tips are not. Replace bands when damaged or they no longer provide enough tension to hold tips tightly in or on ears. Replace tips when they are damaged or are no longer soft and pliable.

[See our Banded Hearing Protectors](#)

Resources for training employees about hearing protection

- [3M Hearing Conservation Worker Training video](#)
- [Roll-Down Foam Earplugs fitting poster \(PDF, 9.10 MB\)](#)

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